

PHOT 301: Quantum Photonics

Problems and final answers of 1.1, 1.2, 1.3, 1.5, 1.8

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Problems

Here we list the problems with their final solutions so you can check whether you have the correct answers. Some problems ask you to prove a theorem, for these problems, I write just some extra hints. The problems are from Griffiths 3rd edition. The problems for this week:

- Chapter 1: 1.1, 1.2, 1.3, 1.5, 1.8

Problem 1.1

For the distribution of ages in the example in Section 1.3.1:

- Compute $\langle j^2 \rangle$ and $\langle j \rangle^2$.
- Determine Δj for each j , and use Equation 1.11 to compute the standard deviation.
- Use your results in (a) and (b) to check Equation 1.12.

Solution (1.1)

- $\langle j^2 \rangle = 459.6$ and $\langle j \rangle^2 = 441$.
- Standard deviation $\sigma = 4.3$.
- $4.3 = \langle (\Delta j)^2 \rangle = \sqrt{18.6}$?

Problem 1.2

- Find the standard deviation of the distribution in Example 1.2.
- What is the probability that a photograph, selected at random, would show a distance x more than one standard deviation away from the average?

Solution (1.2)

- (a) Standard deviation $\sigma = \frac{2}{3\sqrt{5}} h$.
 (b) $1 - \sqrt{\frac{1}{3}(1 + 2/\sqrt{5})} + \sqrt{\frac{1}{3}(1 - 2/\sqrt{5})}$.

Problem 1.3

Consider the gaussian distribution

$$\rho(x) = Ae^{-\lambda(x-a)^2},$$

where A , a , and λ are positive real constants. (The necessary integrals are inside the back cover.)

- (a) Use Equation 1.16 to determine A .
 (b) Find $\langle x \rangle$, $\langle x^2 \rangle$, and σ .
 (c) Sketch the graph of $\rho(x)$.

Solution (1.3)

- (a) $A = \sqrt{\lambda/\pi}$.
 (b) $\langle x \rangle = a$, $\langle x^2 \rangle = \frac{1}{2\lambda} + a^2$, and $\sigma = \frac{1}{\sqrt{2\lambda}}$.
 (c) Bell curve at $x = a$ and maximum $\rho(a) = A$.

Problem 1.5

Consider the wave function

$$\Psi(x, t) = Ae^{-\lambda|x|}e^{-i\omega t},$$

where A , λ , and ω are positive real constants. (We'll see in Chapter 2 for what potential (V) this wave function satisfies the Schrödinger equation.) (a) Normalize Ψ . (b) Determine the expectation values of x and x^2 . (c) Find the standard deviation of x . Sketch the graph of $|\Psi|^2$, as a function of x , and mark the points $(\langle x \rangle + \sigma)$ and $(\langle x \rangle - \sigma)$, to illustrate the sense in which σ represents the “spread” in x . What is the probability that the particle would be found outside this range?

Solution (1.5)

- (a) $A = \sqrt{\lambda}$.
 (b) $\langle x \rangle = 0$, $\langle x^2 \rangle = \frac{1}{2\lambda^2}$.
 (c) $\sigma = \frac{1}{\sqrt{2\lambda}}$. $P(|x| > \sigma) = e^{-\lambda}$.

Problem 1.8

Suppose you add a constant V_0 to the potential energy (by “constant” I mean independent of x as well as t). In classical mechanics this doesn’t change anything, but what about quantum mechanics? Show that the wave function picks up a time-dependent phase factor: $\exp(-iV_0t/\hbar)$. What effect does this have on the expectation value of a dynamical variable?

Solution (1.8)

No effect.